

CS-302: Artificial Intelligence

LECTURE NO 10:

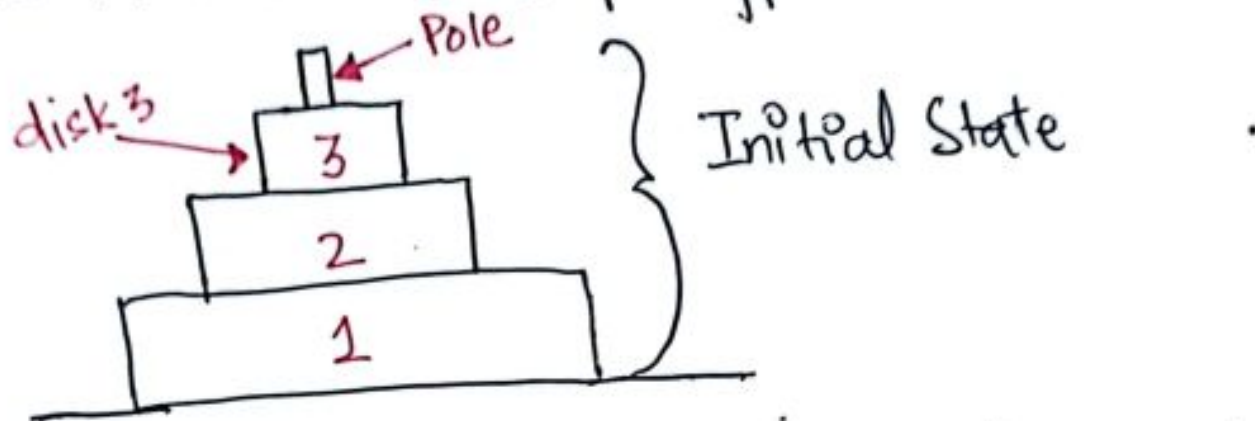
Organization of Lecture

1. Towers of Hanoi Problem
2. Tic-Tac-Toe Problem
3. Propositional Calculus
4. Set of Equivalence Relation or Laws
5. Rules of Inference
6. First Order Logic (FOL)
7. Unification Algorithm in A.I

1. Towers of Hanoi Problem (1/3)

1) Towers of Hanoi Problem:

↳ On one post there is set of 'n' disks each with hole in the center ^{and} of different radius.

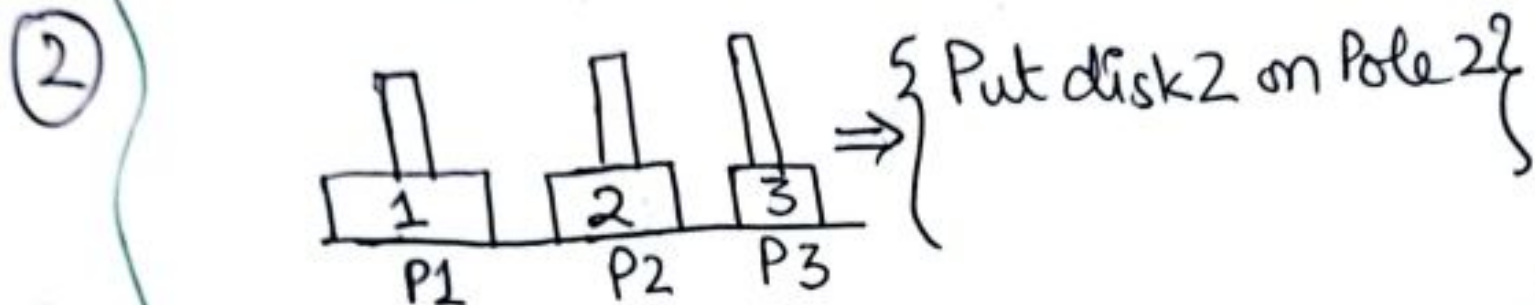
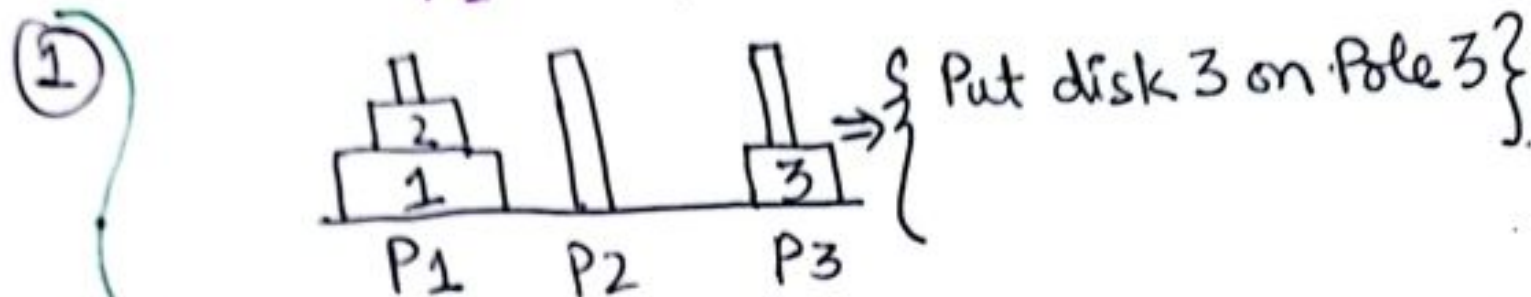


↳ Task is to move all of the disks to one other post.

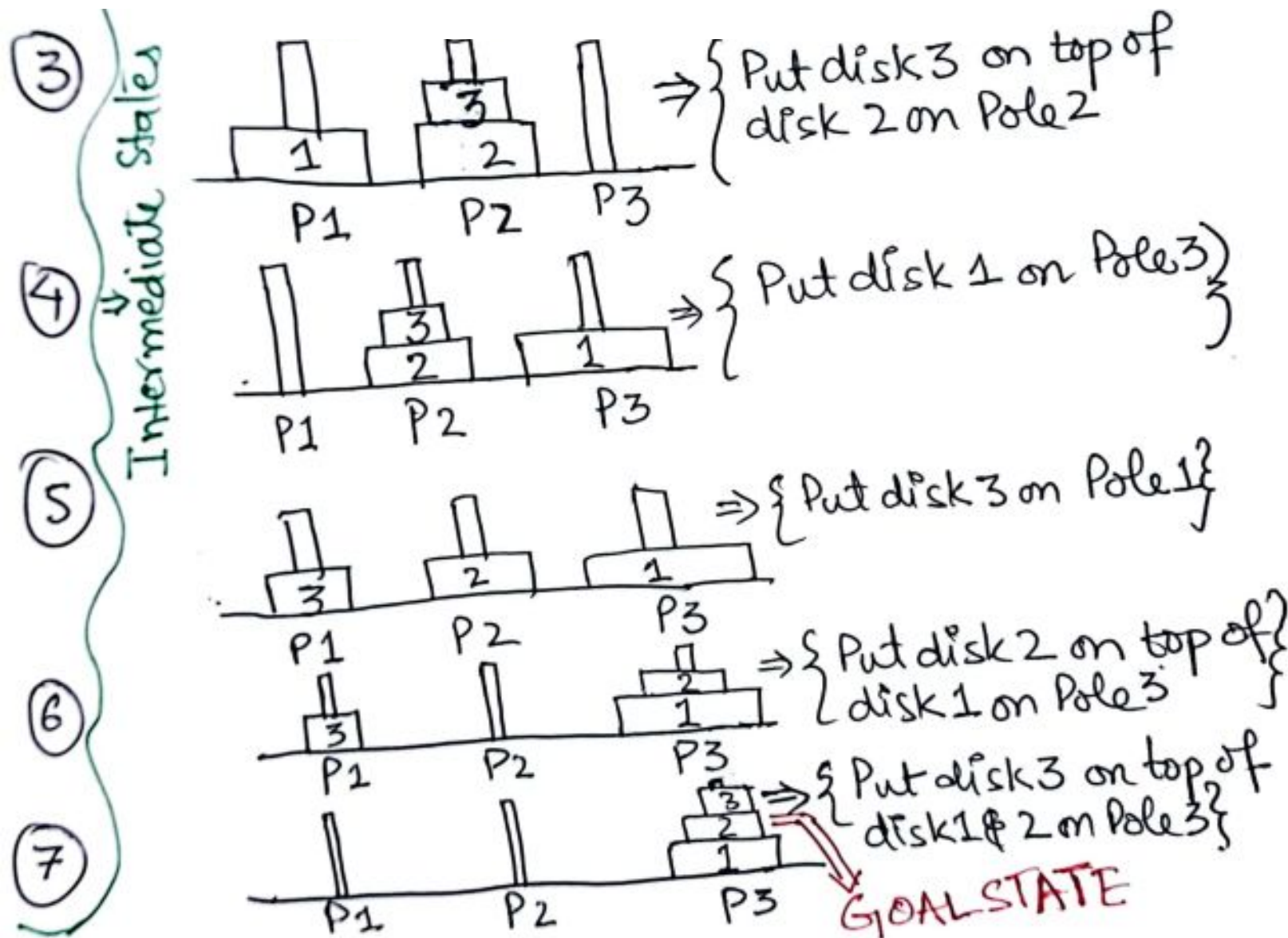
- Only one disk can be moved at a time.
- Large disk can't be placed on smaller disk.

1. Towers of Hanoi Problem (2/3)

Solution:



1. Towers of Hanoi Problem (3/3)

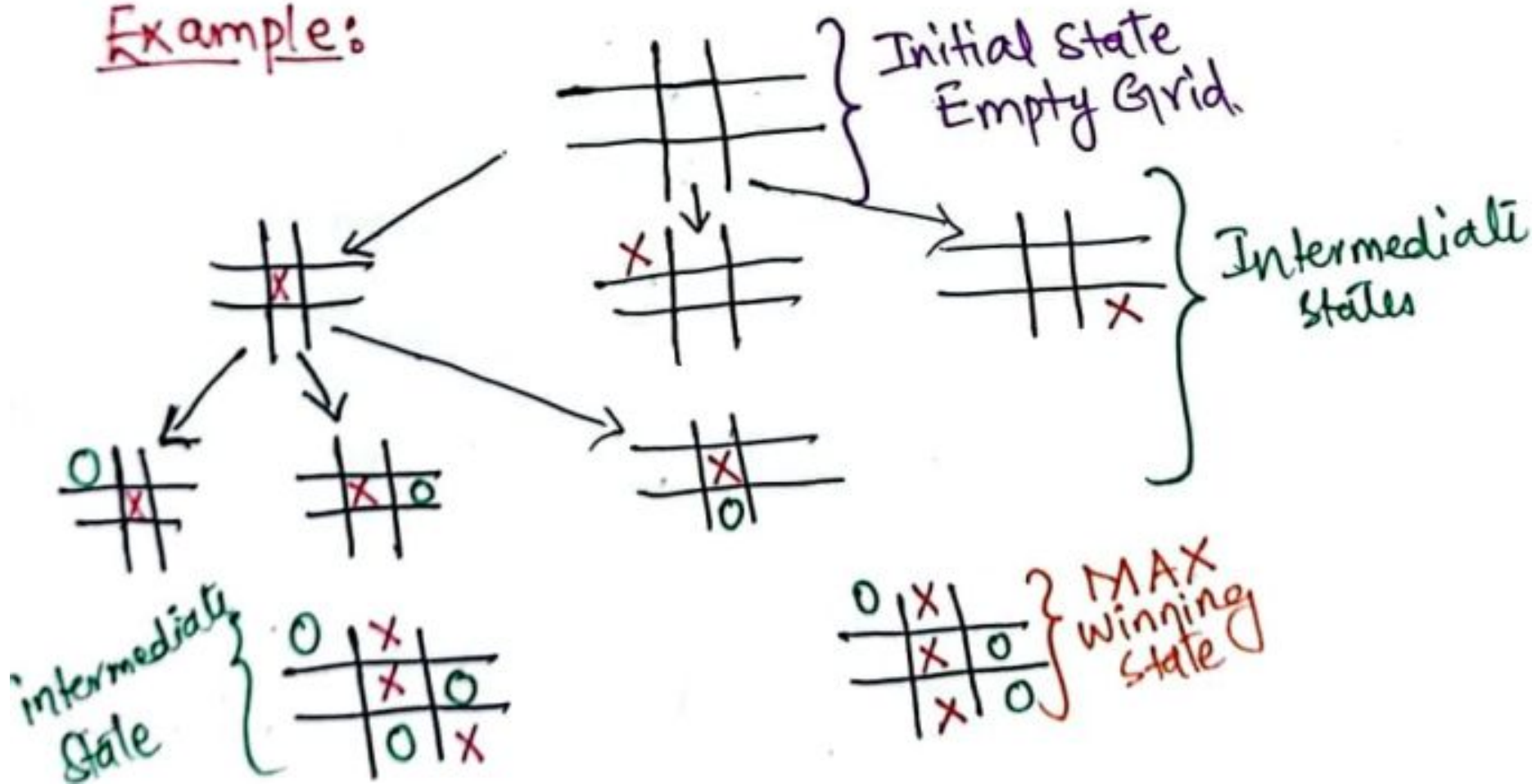


2. Tic-Tac-Toe Problem (1/2)

- Two-Player Game where one player called MAX marks a letter 'X' and the opponent called MIN marks a letter 'O'.
- '3x3' Grid where two players put their letters.
- Player with same marks (three) in complete ROW/COLUMN/DIAGONAL wins the Game.
- Start state = Empty Grid
- Goal state = Win for either of player
 - MAX
(3 consecutive X)
 - MIN
(3 consecutive O).

2. TIC-TAC-TOE Problem (2/2)

Example:



3. Propositional Calculus (1/2)

↳ It is a system that deals with the method used for manipulation of symbols according to some rules.

Alphabet Set:

- i) Set of Variables or Propositional Symbols P, Q, R
- ii) Logical Constants $\begin{matrix} \nearrow \text{True (T)} \\ \searrow \text{False (F)} \end{matrix}$
- iii) Two Parentheses " $($ " " $)$ "
- iv) Set of logical operators

	<u>Word</u>	<u>Symbol</u>	<u>example</u>
↳ a)	not	$\neg$	not of X ($\neg X$)
b)	and	$\wedge$	X and Y ($X \wedge Y$)
c)	OR	$\vee$	X OR Y ($X \vee Y$)
d)	implies	$\rightarrow$	if X then Y ($X \rightarrow Y$)
e)	if & only if	$\leftrightarrow$	$X \leftrightarrow Y$

3. Propositional Calculus (2/2)

Examples:

X: It is Warm

Y: It is Humid.

Z: It is raining.

① If it is humid then it is warm.
'Y' Implies 'X'

$$\hookrightarrow (Y \rightarrow X)$$

② If it is warm and humid then it is not raining.
X ^ Y then not Z

$$(X \wedge Y) \rightarrow \neg Z$$

Propositional Logic is not
used when:
for all, for some

$\hookrightarrow$ for that we use
1st order logic or
predicate logic

4. Set of Equivalence Relation or Laws (1/2)

4) Set of Equivalence Relation or Laws:

(P, Q, R) are Variables

↳ Commutative Laws:

$$P \wedge Q \equiv Q \wedge P, \quad P \vee Q \equiv Q \vee P$$

↳ Associative Laws:

$$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R), \quad (P \vee Q) \vee R \equiv P \vee (Q \vee R)$$

↳ Double Negation:

$$\sim(\sim P) \equiv P$$

↳ De-Morgan's Law:

$$\neg(P \vee Q) = \neg P \wedge \neg Q, \quad \neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

↳ Absorption Law:

$$P \wedge (P \vee Q) \equiv P, \quad P \vee (P \wedge Q) \equiv P$$

4. Set of Equivalence Relation or Laws (2/2)

↳ Law of Contradiction:
 $P \wedge \neg P \equiv \text{False}$ $\left. \begin{array}{l} P=1 \\ \neg P=0 \end{array} \right\} \textcircled{0}$ $\left. \begin{array}{l} P=0 \\ \neg P=1 \end{array} \right\} \textcircled{0}$

↳ Law of excluded middle:
 $P \vee \neg P \equiv \text{True}$ $\left. \begin{array}{l} P=1 \\ \neg P=0 \end{array} \right\} \textcircled{1}$ $\left. \begin{array}{l} P=0 \\ \neg P=1 \end{array} \right\} \textcircled{1}$

↳ Law of idempotency:
 $P \wedge P \equiv P$ $\left[\begin{array}{l} P=1 \rightarrow 1 \\ P=0 \rightarrow 0 \end{array} \right]$

5. Rules of Inference

5) Rules of Inference:

① MODUS PONENS: If ' P ' and ' $P \rightarrow Q$ ' is given to be true, then we can infer that ' Q ' is true.

P : It is a holiday — (T) $\rightarrow$ We can infer that it is true.

Q : the school is closed]

$P \rightarrow Q$: If it is a holiday, then school is closed. — (T)

② MODUS TOLLENS: If ' $\sim Q$ ' and ' $\sim P \rightarrow \sim Q$ ' are given to be true, then we can infer that ' $\sim P$ ' is true.

$\sim Q$: School is not closed — (T)

$\sim P \rightarrow \sim Q$: If it is not a holiday, then school is not closed. — (T)

$\sim P$: It is not a holiday] $\rightarrow$ we can infer that it is true.

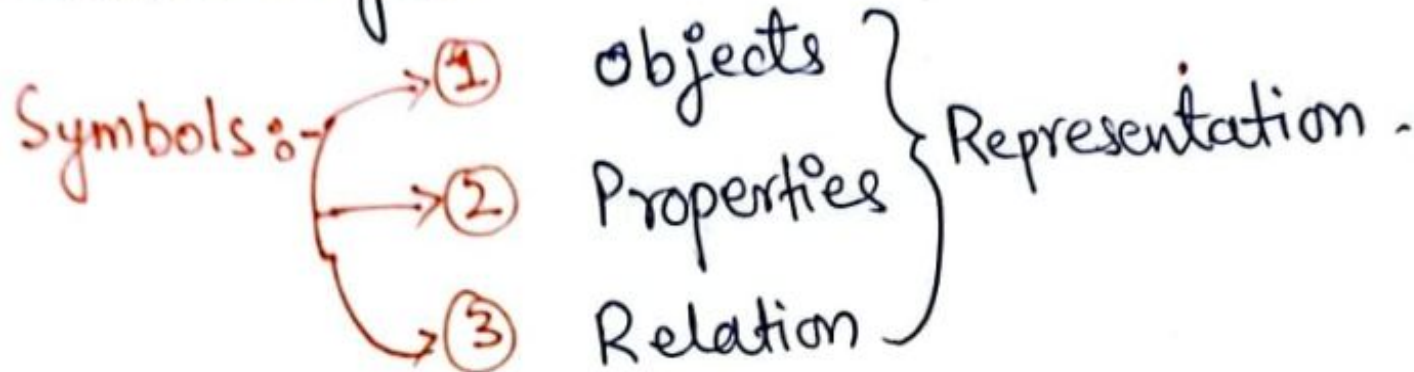
6. First Order Logic (FOL) (1/4)

⑥ First Order Logic (FOL):

FOL = First Order
Predicate Logic

Representing Simple Facts in FOL:

- ↳ Models world in terms of objects.
- ↳ Real-world facts can be represented as logical propositions written as well-formed formulas in propositional logic.



6. First Order Logic (FOL) (2/4)

Symbols are formed of the following:

↳ ① Set of all uppercase english Alphabets.

② Set of digits from 0 to 9.

③ Underscore character

[All dogs are brown]
↳ Written in FOL

→ Can't be written in propositional logic

Defining a Sentence:

→ Every atomic sentence is a sentence.

→ Atomic sentence is defined as predicate constant of arity 'n', followed by $t_1, t_2, \dots, t_n$ terms enclosed in parentheses and separated by commas.

① If 'S' is sentence then $\neg S$ is (also) sentence.

6. First Order Logic (FOL) (3/4)

- ② If S_1, S_2 are sentences then $S_1 \wedge S_2$ is also a sentence.] *Conjunction*
- ③ If S_1, S_2 are sentences then $S_1 \vee S_2$ is also a sentence.] *Disjunction*
- ④ If S_1, S_2 are sentences then $S_1 \rightarrow S_2$ is also a sentence.] *Implication*
- ⑤ If S_1, S_2 are sentences then $S_1 \equiv S_2$ is also a sentence.] *Equivalence*
- ⑥ If x is a variable and S is sentence then
for all x , $\forall x S$ is a sentence.] *Universal Quantifier*
- ⑦ If x is a variable and S is sentence then
 $\exists x S$ is a sentence.] *Existential Quantifier*.
there exists some x

6. First Order Logic (FOL) (4/4)

Quantifiers in Predicate Calculus:

There are two Quantifiers used in FOPL

① Universal Quantifier ($\forall x$): for all 'x' such that

② Existential Quantifier ($\exists x$): for some 'x' such that

↳ constrain the meaning of sentence containing a variable

↳ Quantifier is followed by a variable & a sentence.

Example:

① All boys like football $\rightarrow \forall x: \text{Boys}(x) \Rightarrow \text{Like}(x, \text{football})$

② Some boys like football $\rightarrow \exists x: \text{Boy}(x) \wedge \text{Like}(x, \text{football})$

7. Unification Algorithm in A.I (1/4)

↳ Unification means making expressions look identical.

↳ It can be done with the process of substitution.

Simple Eg:-

$$p(x, F(y)) - \textcircled{1} , p(a, F(g(z))) - \textcircled{2}$$

↳ $\textcircled{1}$ and $\textcircled{2}$ are identical if:

$$x = a \ \& \ y = g(z)$$

So eq $\textcircled{1}$ becomes

$$p(a, F(g(z))) - \textcircled{1}$$

Now eq $\textcircled{1}$ & $\textcircled{2}$ are identical $\Rightarrow$ unification.

Unification:

$\leftarrow$ 'x' is replaced with 'a'

Substitution Set $\Rightarrow [a/x, g(z)/y]$ $\rightarrow$ 'y' is replaced with 'g(z)'.

7. Unification Algorithm in A.I (2/4)

Unification conditions:-

- ① Predicate Symbol must be same.
↳ In above example we use 'P' in both equation.
 ↳ is predicate
- ② No. of arguments in both expressions must be identical.
↳ In above example we have 2 arguments in both expressions.
- ③ If two similar variables present in same expression then unification fails.

7. Unification Algorithm in A.I (3/4)

Algorithm:

Unify(A1, A2)

① If A1 or A2 is variable/constant

↳ if A1 and A2 are identical
return NIL

↳ Else if A1 occurs in A2 return fail

↳ Else return A2/A1

↳ Check for A2 in A1

↳ fail if A2 occurs in A1

↳ Else return A1/A2

② if predicate not same] Fail

③ if different arguments] Fail

④ Else SUBST to NIL

⑤ Loop

⑥ Return SUBST

7. Unification Algorithm in A.I (4/4)

Question:- $\underbrace{Q(a, g(x, a), f(y))}_{A1}, \underbrace{Q(a, g(f(b), a), x)}_{A2}$

→ substitute "x" with "f(b)" i.e., $[f(b)/x]$

→ $Q(a, g(f(b), a), f(y)), Q(a, g(f(b), a), f(b))$

→ substitute "b" with "y" i.e. $[b/y]$

→ $\underbrace{Q(a, g(f(b), a), f(b))}_{A1}, \underbrace{Q(a, g(f(b), a), f(b))}_{A2}$

→ unified successfully. $[A1 = A2]$

References:

Textbooks

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3. *"Computational Intelligence: A logical Approach"*, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.
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6. *"Artificial Intelligence: Theory and Practice"* by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.
7. *Related documents from open source, mainly internet.*